

Probability & Random Processes

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Department of Electrical Engineering

2.3

Counting Sample Points

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
SCIENCES, LAHORE CAMPUS



Definition 2.1:

The set of all possible outcomes of a statistical experiment is called the **sample space** and is represented by the symbol S .

Definition 2.2:

An **event** is a subset of a sample space.

Definition 2.3:

The **complement** of an event A with respect to S is the subset of all elements of S that are not in A . We denote the complement of A by the symbol A' .

Definition 2.4:

The **intersection** of two events A and B , denoted by the symbol $A \cap B$, is the event containing all elements that are common to A and B .

Definition 2.5:

Two events A and B are **mutually exclusive**, or **disjoint**, if $A \cap B = \phi$, that is, if A and B have no elements in common.

Definition 2.6:

The **union** of the two events A and B , denoted by the symbol $A \cup B$, is the event containing all the elements that belong to A or B or both.



2.3 Counting Sample Points

- Element of chance associated with the occurrence of certain events
- Solve a probability problem by counting the number of points in the sample space without actually listing each element.
- Fundamental principle of counting, often referred to as the **multiplication rule**

Rule 2.1: If an operation can be performed in n_1 ways, and if for each of these ways a second operation can be performed in n_2 ways, then the two operations can be performed together in $n_1 n_2$ ways.

Example:

How many sample points are there in the sample space when a pair of dice is thrown once?

The first die can land face-up in any one of $n_1 = 6$ ways. For each of these 6 ways, the second die can also land face-up in $n_2 = 6$ ways. Therefore, the pair of dice can land in $n_1 n_2 = (6)(6) = 36$ possible ways.



Example:

A developer of a new subdivision offers prospective home buyers a choice of Tudor, rustic, colonial, and traditional exterior styling in ranch, two-story, and split-level floor plans.

In how many different ways can a buyer order one of these homes?

Solution:

Since $n_1 = 4$ and $n_2 = 3$, a buyer must choose from

$$n_1 n_2 = (4)(3) = 12 \text{ possible homes.}$$

Lets construct a tree diagrams and counting the various paths along the branches.

There will be $n_1 = 4$ branches corresponding to the different exterior styles, and then there will be $n_2 = 3$ branches extending from each of these 4 branches to represent the different floor plans.

This tree diagram yields the $n_1 n_2 = 12$ choices of homes given by the paths along the branches.



Example:

If a 22-member club needs to elect a chair and a treasurer, how many different ways can these two to be elected?

For the chair position, there are 22 total possibilities. For each of those 22 possibilities, there are 21 possibilities to elect the treasurer. Using the multiplication rule, we obtain

$$n_1 \times n_2 = 22 \times 21 = 462 \text{ different ways.}$$

Suppose customer wishes to buy a new cell phone and can choose from 5 brands, 5 sets of capability, and 4 colors.

These three classifications result in $n_1 n_2 n_3 = (5)(5)(4) = 100$ different ways for a customer to order one of these phones.

Generalized multiplication rule covering k operations:**Rule 2.2:**

If an operation can be performed in n_1 ways, and if for each of these a second operation can be performed in n_2 ways, and for each of the first two a third operation can be performed in n_3 ways, and so forth, then the sequence of k operations can be performed in $n_1 n_2 \cdots n_k$ ways.



Example:

Sam is going to assemble a computer by himself. He has the choice of chips from two brands, a hard drive from four, memory from three, and an accessory bundle from five local stores. How many different ways can Sam order the parts?

Solution:

Since $n_1 = 2$, $n_2 = 4$, $n_3 = 3$, and $n_4 = 5$
there are $n_1 \times n_2 \times n_3 \times n_4 = 2 \times 4 \times 3 \times 5$
 $= 120$
different ways to order the parts.

Definition 2.7: A **permutation** is an arrangement of all or part of a set of objects.



Example:

Consider the three letters a , b , and c . The possible permutations are abc , acb , bac , bca , cab , and cba . Thus, we see that there are 6 distinct arrangements.

There are $n_1 = 3$ choices for the first position.

No matter which letter is chosen, there are always $n_2 = 2$ choices for the second position.

No matter which two letters are chosen for the first two positions, there is only $n_3 = 1$ choice for the last position,

giving a total of $n_1 n_2 n_3 = (3)(2)(1) = 6$ permutations

by Rule 2.2.

In general, n distinct objects can be arranged in

$n(n-1)(n-2) \cdots (3)(2)(1)$ ways.

notation for such a number.

Definition 2.8: For any non-negative integer n , $n!$, called “ n factorial,” is defined as

$$n! = n(n-1) \cdots (2)(1),$$

with special case $0! = 1$.



Theorem 2.1: The number of permutations of n objects is $n!$.

Consider the number of permutations that are possible by taking two letters at a time from four.

These would be $ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db$, and dc .

Using Rule 2.1 again, we have two positions to fill, with $n_1 = 4$ choices for the first and then $n_2 = 3$ choices for the second, for a total of $n_1 n_2 = (4)(3) = 12$ permutations.

In general, n distinct objects taken r at a time can be arranged in

$$n(n-1)(n-2) \cdots (n-r+1)$$

Theorem 2.2: The number of permutations of n distinct objects taken r at a time is

$${}_n P_r = \frac{n!}{(n-r)!}.$$



Example:

In one year, three awards (research, teaching, and service) will be given to a class of 25 graduate students in a statistics department. If each student can receive at most one award, how many possible selections are there?

Solution:

Since the awards are distinguishable, it is a permutation problem

Total number of sample points is:

$${}_{25}P_3 = \frac{25!}{(25-3)!} = \frac{25!}{22!} = (25)(24)(23) = 13,800.$$



Example:

A president and a treasurer are to be chosen from a student club consisting of 50 people. How many different choices of officers are possible if

- (a) there are no restrictions;
- (b) *A* will serve only if he is president;
- (c) *B* and *C* will serve together or not at all;
- (d) *D* and *E* will not serve together?

Solution:

- a) The total number of choices of officers, without any restrictions, is

$${}_{50}P_2 = \frac{50!}{48!} = (50)(49) = 2450.$$

b) Since *A* will serve only if he is president, we have two situations here: (i) *A* is selected as the president, which yields 49 possible outcomes for the treasurer's position, or (ii) officers are selected from the remaining 49 people without *A*, which has the number of choices ${}_{49}P_2 = (49)(48) = 2352$. Therefore, the total number of choices is $49 + 2352 = 2401$.

c) The number of selections when *B* and *C* serve together is 2. The number of selections when both *B* and *C* are not chosen is ${}_{48}P_2 = 2256$. Therefore, the total number of choices in this situation is $2 + 2256 = 2258$.



d) The number of selections when D serves as an officer but not E is $(2)(48) = 96$, where 2 is the number of positions D can take and 48 is the number of selections of the other officer from the remaining people in the club except E .

The number of selections when E serves as an officer but not D is also $(2)(48) = 96$.

The number of selections when both D and E are not chosen is ${}_{48}P_2 = 2256$.

Therefore, the total number of choices is $(2)(96) + 2256 = 2448$.



Theorem 2.3: The number of permutations of n objects arranged in a circle is $(n - 1)!$.

So far we have considered permutations of distinct objects.

That is, all the objects were completely different or distinguishable.

Obviously, if the letters b and c are both equal to x , then the 6 permutations of the letters a , b , and c become

axx , axx , xax , xax , xxa , and xxa , of which only 3 are distinct.

Therefore, with 3 letters, 2 being the same, we have $3!/2! = 3$ distinct permutations.

With 4 different letters a , b , c , and d , we have 24 distinct permutations. If we let $a = b = x$ and $c = d = y$, we can list only the following distinct permutations: $xyyy$, $xyxy$, $yxyx$, $yyxx$, $xyyx$, and $yxyx$. Thus, we have $4!/(2! 2!) = 6$ distinct permutations

Theorem 2.4: The number of distinct permutations of n things of which n_1 are of one kind, n_2 of a second kind, $\dots$, n_k of a k th kind is

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$



Example:

In a college football training session, the defensive coordinator needs to have 10 players standing in a row.

Among these 10 players, there are 1 freshman, 2 sophomores, 4 juniors, and 3 seniors. How many different ways can they be arranged in a row if only their class level will be distinguished?

Total number of arrangements is

$$\frac{10!}{1! 2! 4! 3!} = 12,600.$$



Sometimes we are concerned with the number of ways of partitioning a set of n objects into r subsets called **cells**.

A partition has been achieved if the intersection of every possible pair of the r subsets is the empty set $\varnothing$ and if the union of all subsets gives the original set.

The order of the elements within a cell is of no importance.

Consider the set $\{a, e, i, o, u\}$.

The possible partitions into two cells in which the first cell contains 4 elements and the second cell 1 element are:

$$\{(a, e, i, o), (u)\}, \{(a, i, o, u), (e)\}, \{(e, i, o, u), (a)\}, \{(a, e, o, u), (i)\}, \{(a, e, i, u), (o)\}.$$

there are 5 ways to partition a set of 4 elements into two subsets, or cells, containing 4 elements in the first cell and 1 element in the second.

$$\binom{5}{4, 1} = \frac{5!}{4! 1!} = 5,$$



Theorem 2.5: The number of ways of partitioning a set of n objects into r cells with n_1 elements in the first cell, n_2 elements in the second, and so forth, is

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!},$$

where $n_1 + n_2 + \cdots + n_r = n$.

we are interested in the number of ways of selecting r objects from n without regard to order. These selections are called **combinations**

A combination is actually a partition with two cells, the one cell containing the r objects selected and the other cell containing the $(n-r)$ objects that are left.

$$\binom{n}{r, n-r}, \text{ is usually shortened to } \binom{n}{r}$$

the number of elements in the second cell must be $n - r$.

